NLP

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
df = pd.read_csv('train.txt', sep = ';', header = None, names = ['text', 'em
```

```
df.head()
```

	text	emotion
0	i didnt feel humiliated	sadness
1	i can go from feeling so hopeless to so damned...	sadness
2	im grabbing a minute to post i feel greedy wrong	anger
3	i am ever feeling nostalgic about the fireplac...	love
4	i am feeling grouchy	anger

Next steps: Generate code with df | New interactive sheet

```
df.isnull().sum()
```

	0
text	0
emotion	0

dtype: int64

```
unique_emotion = df['emotion'].unique()
emotion_num = {}
i = 0
for emo in unique_emotion:
  emotion_num[emo] = i
  i += 1
df['emotion'] = df['emotion'].map(emotion_num)
```

```
df
```

	text	emotion
0	i didnt feel humiliated	0
1	i can go from feeling so hopeless to so damned...	0
2	im grabbing a minute to post i feel greedy wrong	1
3	i am ever feeling nostalgic about the fireplac...	2
4	i am feeling grouchy	1
...	...	...
15995	i just had a very brief time in the beanbag an...	0
15996	i am now turning and i feel pathetic that i am...	0
15997	i feel strong and good overall	5
15998	i feel like this was such a rude comment and i...	1
15999	i know a lot but i feel so stupid because i ca...	0

16000 rows × 2 columns

Next steps: Generate code with df | New interactive sheet

```
df['text'] = df['text'].apply(lambda x : x.lower())
```

```
import string

def  remove_punc(txt):
  return txt.translate(str.maketrans('', '', string.punctuation))
```

```
string.punctuation # this is the list of punctuation in string library which
```

```
'!"#$%&\'()*+,-./:;<=>?@[\\]^_`{|}~'
```

```
df['text'] = df['text'].apply(remove_punc)
```

```
def remove_num(txt):
  new =""
  for i in txt:
    if not i.isdigit():
      new = new + i
  return new
df['text'] = df['text'].apply(remove_num)
```

```
def remove_emojis(txt):
    new = ""
    for i in txt:
        if i.isascii():
```

```
            new += i
    return new

df['text'] = df['text'].apply(remove_emojis)
```

```
import nltk
```

```
from nltk.corpus import stopwords
from nltk.tokenize import word_tokenize
```

```
nltk.download('punkt')
nltk.download('stopwords')
```

```
[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data]   Unzipping tokenizers/punkt.zip.
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data]   Unzipping corpora/stopwords.zip.
True
```

```
stop_words = set(stopwords.words('english'))
```

```
df.loc[1]['text']
```

```
'i can go from feeling so hopeless to so damned hopeful just from being arou
nd someone who cares and is awake'
```

```
def remove(txt):
  words = txt.split()
  cleaned = []
  for i in words:
    if not i in stop_words:
      cleaned.append(i)

  return ' '.join(cleaned)
```

```
df['text'] = df['text'].apply(remove)
```

```
df.loc[1]['text']
```

```
'go feeling hopeless damned hopeful around someone cares awake'
```

```
df.head()
```

	text	emotion
0	didnt feel humiliated	0
1	go feeling hopeless damned hopeful around some...	0
2	im grabbing minute post feel greedy wrong	1
3	ever feeling nostalgic fireplace know still pr...	2
4	feeling grouchy	1

Next steps: Generate code with df | New interactive sheet

```
import nltk
nltk.download('punkt_tab')
df['tokens'] = df['text'].apply(word_tokenize)

df.head()
```

```
[nltk_data] Downloading package punkt_tab to /root/nltk_data...
[nltk_data]   Unzipping tokenizers/punkt_tab.zip.
```

	text	emotion	tokens
0	didnt feel humiliated	0	[didnt, feel, humiliated]
1	go feeling hopeless damned hopeful around some...	0	[go, feeling, hopeless, damned, hopeful, aroun...
2	im grabbing minute post feel greedy wrong	1	[im, grabbing, minute, post, feel, greedy, wrong]
3	ever feeling nostalgic fireplace know still pr...	2	[ever, feeling, nostalgic, fireplace, know, st...
4	feeling grouchy	1	[feeling, grouchy]

Next steps: Generate code with df | New interactive sheet

```
df.loc[1]['tokens']
```

```
['go',
 'feeling',
 'hopeless',
 'damned',
 'hopeful',
 'around',
 'someone',
 'cares',
 'awake']
```

```
from nltk.stem import WordNetLemmatizer
nltk.download('wordnet')
nltk.download('omw-1.4')
```

```
[nltk_data] Downloading package wordnet to /root/nltk_data...
[nltk_data] Downloading package omw-1.4 to /root/nltk_data...
True
```

```
lemmatizer = WordNetLemmatizer()
```

```
def lemmatize_words(words):
    return [lemmatizer.lemmatize(word) for word in words]

df['lemmatized'] = df['tokens'].apply(lemmatize_words)
```

```
df.loc[1]['lemmatized']
```

```
['go',
 'feeling',
 'hopeless',
 'damned',
 'hopeful',
 'around',
 'someone',
 'care',
 'awake']
```

```
from nltk.stem import PorterStemmer

stemmer = PorterStemmer()

def stem_words(words):
    return [stemmer.stem(word) for word in words]

df['stemmed'] = df['tokens'].apply(stem_words)

df.loc[1]['stemmed']
```

```
['go', 'feel', 'hopeless', 'damn', 'hope', 'around', 'someon', 'care',
'awak']
```

```
Start coding or generate with AI.
```